

$$\begin{cases} \partial_t f = \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) = f_0 \end{cases}$$

- What happens as $t \rightarrow \infty$?

• Part 1 (Rakotomalala & W, in review, 2025):

o Existence of compact global attractor

■ Robinson, *Infinite-Dimensional Dynamical Systems*

9 Attractor has dimension at least 4 if and only if it is linearly unstable



Antdynamics

Ant dynamics

$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) &= f_0 \end{cases}$$

- What happens as $t \rightarrow \infty$?
- Part 1 (Rakotomalala & dW, in review, 2025):
 - Existence of compact global attractor
 - Robinson, *Infinite-Dimensional Dynamical Systems*
 - Attractor has dimensional lower bound of 4 if homogeneous state is linearly unstable



Ant dynamics

$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) &= f_0 \end{cases}$$

- What happens as $t \rightarrow \infty$?
- Part 2 (Rakotomalala & dW, in review, 2026):

